October 15, 1958

MEMO TO: Mr. S. W. Dunwell

SUBJECT: Information to be discussed at Los Alamos

In accord with our agreement last week, I intend to discuss the following items with members of the Mathematical Planning Group at Los Alamos when I am there October 16-18.

1. Progress Report on the Data Flow Model

   The present results confirm experimentally the previous theoretical speed estimates for floating operations of 1.0 us add, 1.95 us mpy, and 7.0 us div.

   The data flow model is presently executing the 8 bit path thru the adder in 0.45 us, which is less than the time originally estimated.

   The double transistor card version now under design will have an indirect beneficial effect on speed by making the units more compact.

2. The half microsecond memory

   At the present time indications are that IBM may not include the half microsecond memory boxes in its commercial STRETCH program for economic reasons. The memories will be built to satisfy contractural obligations. Perhaps, however, LASL might want to consider substituting regular 2 microsecond memory boxes, for the half microsecond boxes if it proved advantageous to them.

3. Recent Instruction changes
   - (a) The changes in the AUGMENT instructions.
   - (b) 48 bit limit on VFL Multiply and Divide.
   - (c) Their feeling concerning offset 20 in VFL Multiply and Divide.

4. Programming

   Discuss progress of joint LASL-Applied Programming effort.

5. Test Problem
   Discuss present plans and obtain schedule, if possible.

6. Discuss planning letter sent to BuStandards concerning power equipment ,etc.

7. Discuss Westinghouse proposal configuration if questions arise.

8. Discuss 709TX if questions arise.

9. Question them concerning on-line punch requirements

   (a) Their feeling as to the necessity of on-line punching.
   (b) What is the present punch use on 704's?
   (c) How sensitive are these uses to speed of punch?
   (d) What is "best" speed for STRETCH?

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cc: Mr. D. W. Sweeney
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